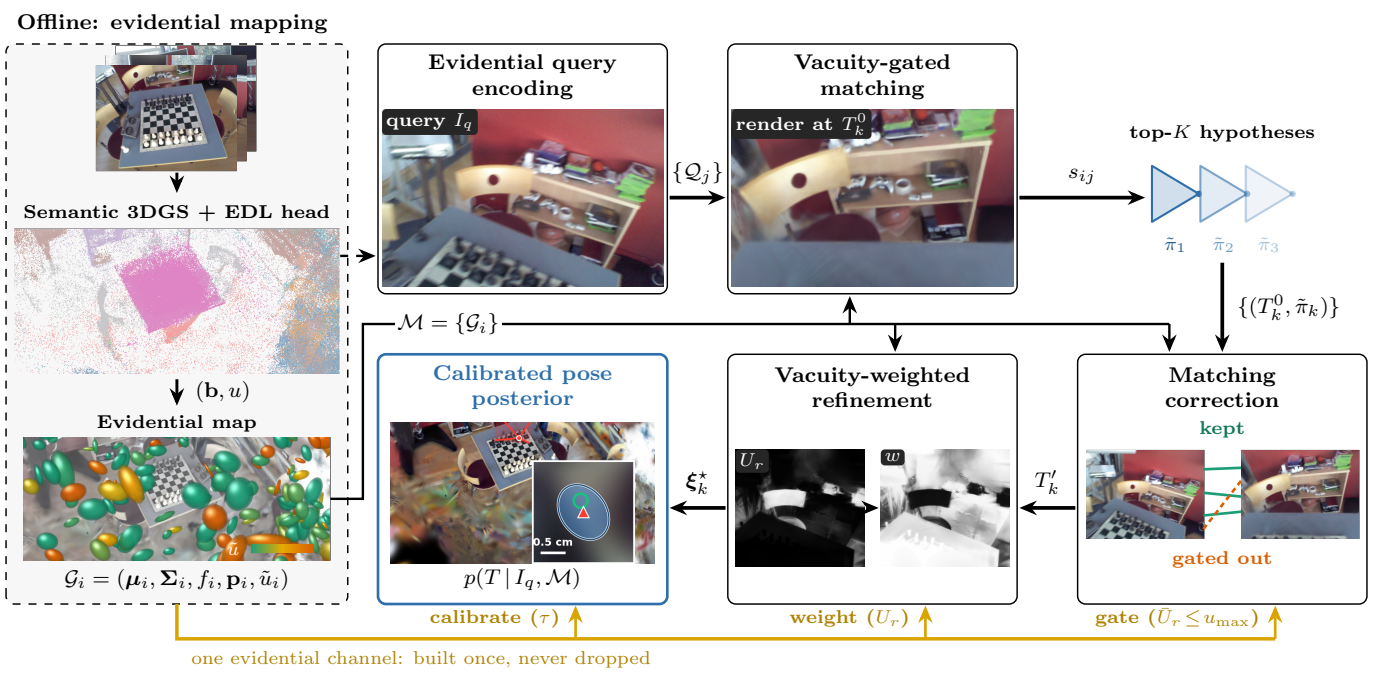

Offline: evidential mapping
Semantic 3DGS + EDL head
$(\mathbf{b}, u)$
Evidential map
$\tilde{u}$
$\mathcal{G}_i = (\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, f_i, \mathbf{p}_i, \tilde{u}_i)$
Evidential query encoding
query I_q
$\{\mathcal{Q}_j\}$
Vacuity-gated matching
render at T_k^0
s_{ij}
top-K hypotheses
$\tilde{\pi}_1$ $\tilde{\pi}_2$ $\tilde{\pi}_3$
$\{(T_k^0, \tilde{\pi}_k)\}$
$\mathcal{M} = \{\mathcal{G}_i\}$
Calibrated pose posterior
0.5 cm
$p(T \mid I_q, \mathcal{M})$
$\boldsymbol{\xi}_k^\star$
Vacuity-weighted refinement
U_r
w
T_k'
Matching correction
kept
gated out
calibrate (τ)
weight (U_r)
gate ($\bar{U}_r \le u_{\max}$)
one evidential channel: built once, never dropped